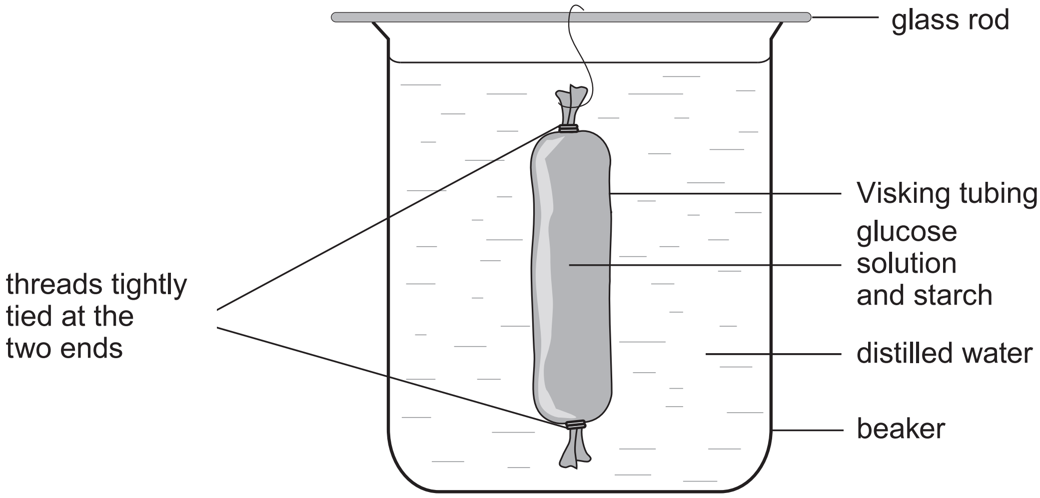
Examiner only

Answer **all** questions.

1. A student set up the following experiment to show how diffusion of digested foods occurs in the small intestine. The Visking tubing contained starch and a solution of glucose with a concentration of 100 g/dm^3. This was placed in a beaker containing distilled water at 20 °C and left for 2 hours.



(a) (i) Explain what is expected to happen to the starch and the glucose in the Visking tubing during the experiment. [4]

....

....

....

....

....

....

(ii) Describe **two** chemical tests the student should carry out on the contents of the beaker to confirm your answer to (a)(i) and state the expected results. [4]

1. ....

....

....

2. ....

....

....

 (3445UA0-1)

Examiner only

(b) Explain how each of the following changes would affect the diffusion process in this experiment.

(i) Carbohydrase is added to the contents of the Visking tubing. [2]

....

....

....

(ii) The investigation is carried out at 35 °C. [2]

....

....

....

(iii) The experiment is carried out with a solution of glucose of 100 g/dm^3 concentration replacing the distilled water in the beaker. [2]

....

....

....

(c) State **one** of the limitations of using Visking tubing as a model intestine. [1]

....

....

(d) Explain **two** adaptations of the small intestine for the absorption of digested food. [4]

1. ....

....

....

2. ....

....

....

3445UA01 03

19